Name: _____ **Period:** _____ **Date:** _____

Topic 2.8 Guided Notes: Intelligence and Achievement

Unit 2: Cognition | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic2.8

Word Bank

Mental Age	*Intelligence Quotient (IQ)*	*Intelligence*
Aptitude Tests	*Achievement Tests*	*General Intelligence (g)*
Crystallized Intelligence	*Fluid Intelligence*	*Chronological Age*

📺 *Video Notes: "The Controversy of Intelligence" (Crash Course Psychology #23)*

Watch the video above, then answer each question below in your own words.

1. What did Spearman propose, what statistical procedure did he help develop, and how did Thurstone challenge his idea?

2. What evidence does Gardner use for multiple intelligences, how does Sternberg's view differ, and what does the video say research suggests about both?

3. Why did Binet and Simon create their test, what did Binet believe about whether intelligence is fixed, and what did he fear would happen to the test?

4. The video works out IQ for Bruno (age 6, mental age 6) and his sister Betty (age 4, mental age 5). How is IQ calculated in this original formula, what are their scores, and why does the formula break down for adults?

5. What is eugenics, who coined the term, and how does the video say intelligence tests were used in the U.S. and in Nazi Germany?

1. The Nature of Intelligence

At its core, _____ is generally defined as the ability to learn from experience, reason, solve problems, and adapt to new situations. However, theories on how it works fall into two main camps:

- _____: Proposed by Charles Spearman using a statistical method called factor analysis, this theory argues that a single, underlying general mental ability fuels performance on all cognitive tasks. If you are naturally good at math, you are likely naturally good at reading, too.
- Fluid vs. Crystallized Intelligence: Raymond Cattell broke intelligence down into two types. _____ is your ability to think quickly, solve new problems, and abstractly reason (which peaks in young adulthood and slowly declines). _____ is your accumulated knowledge and vocabulary (which continues to grow as you age).

Apply It: Fluid or Crystallized?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A 19-year-old college freshman breezes through a brand-new Sudoku puzzle she's never seen before, quickly spotting patterns and working out the logic on the fly.	
2	A retired librarian in her eighties can recall the exact publication year and author of hundreds of classic novels she has read over her lifetime.	

2. The Metrics of the Mind: IQ and Standardized Testing

Early attempts to measure intelligence relied on the concept of an _____. Originally, IQ was calculated by taking a child's _____ (the level of performance typical of a certain age group) and dividing it by their _____ (their actual age in years), then multiplying by 100.

Today, tests are much more sophisticated. They are given in two main formats:

- _____: Designed to predict your future ability or capacity to learn a new skill (e.g., the SAT or an armed services vocational test).
- _____: Designed to reflect what you have already learned or mastered in a specific subject (e.g., your AP Psychology Exam!).

Apply It: Aptitude or Achievement?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A hiring manager gives job applicants a mechanical-reasoning test to gauge how well they might learn to operate brand-new machinery they've never used before.	
2	A driving instructor gives a written test on the state's traffic laws that students have spent the past month studying in class.	

Word Bank

Standardization	*Reliability*	*Split-Half*	*Predictive Validity*
Norming	*Validity*	*Stereotype Threat*	*Stereotype Lift*
Growth Mindset	*Construct Validity*	*Fixed Mindset*	*Flynn Effect*
Test-Retest			

3. The Golden Rules of Psychometrics

For any psychological assessment to be considered scientifically useful and ethical, it must adhere to three critical psychometric principles:

- _____: Administering and scoring the test in the exact same, consistent way for all test takers.
- This is measured through _____, which is establishing standards by comparing scores to a representative sample population (creating a bell curve).
- _____: The test produces stable and consistent results over time.
- We measure reliability by utilizing methods such as _____ (giving the same test twice) or _____ (comparing scores on two halves of the test).
- _____: The test accurately measures exactly what it is intended to measure.
- Validity can be measured by evaluating _____ (does it test the right concept?) and _____ (does it accurately predict future performance?).

Apply It: Which Psychometric Principle?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	Every student taking the AP Exam receives the same instructions, the same time limit, and has their essays scored using the same official rubric.	
2	A personality test is given to job candidates, and when the same candidates retake it two weeks later, they score almost identically both times.	
3	College admissions officers find that students who scored higher on the SAT years ago also went on to earn notably higher freshman-year GPAs.	

4. Systemic Issues, Bias, and Mindset

Intelligence testing has a dark history. Historically, IQ scores were weaponized to limit access to jobs, military ranks, and immigration. Even today, tests can reflect deep sociocultural biases. Poverty, discrimination, and lack of access to quality education heavily impact scores.

- _____: The global, observed rise in average IQ scores over generations. Humans aren't necessarily genetically mutating to be smarter; rather, we have better nutrition, healthcare, and educational access than our ancestors did.

- _____ occurs when someone risks underperforming on a test because they are highly aware of a negative stereotype regarding their group. _____ is the opposite: performance improves when individuals are aware of positive stereotypes about their group.
- Coined by Carol Dweck, a _____ is the belief that intelligence is static and unchangeable. A _____ is the belief that intelligence can be developed through effort, failure, and persistence—which directly leads to higher academic achievement!

Apply It: Threat, Lift, or Mindset?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A soccer player tells her teammates that talent is something you're either born with or not, and that extra effort in practice won't really change her skill level.	
2	Right before a big math exam, a teacher off-handedly reminds a group of students about a stereotype claiming their gender underperforms in math, and several of those students go on to score below their usual level.	

📺 *Video Notes: "Brains vs. Bias" (Crash Course Psychology #24)*

Watch the video above, then answer each question below in your own words.

1. What are the WAIS and WISC, are they aptitude or achievement tests, and what does the video say these tests are most effectively used for?

2. What three marks must a test hit to be widely accepted, and how does the video say each one is established or checked?

3. What evidence from twin and adoption studies does the video give that genes influence intelligence?

4. What did J. McVicker Hunt find in the Iranian orphanage, and what does it show about the role of environment?

5. What makes a test biased? The video describes bias coming from the test's questions, from who gives the test, and from the test-takers' own expectations — give an example of each, and name the researchers credited with identifying stereotype threat.

5. Don't Trip Up! (Common Misconceptions)

⚠️ Reliability vs. Validity: You can have a highly reliable test that is completely invalid. For example, if you measure intelligence using a tape measure to find the circumference of someone's head, the test has high _____ (you will get the exact same measurement every time), but it has no _____, because head size does not measure intelligence!

⚠️ Aptitude vs. Achievement: _____ point to the future (they predict your capacity to learn). _____ point to the past (they assess what you've already learned).

6. Quick Check: Review Questions

Circle the best answer for each question below.

1. A 70-year-old retired professor cannot solve brand new abstract logic puzzles as quickly as he once could in his youth, but he possesses an enormous vocabulary and a vast, accumulated knowledge base built over his entire lifetime that continues to grow. This type of intelligence is called:

- **A.** Crystallized Intelligence
- **B.** Stereotype Threat
- **C.** Fluid Intelligence
- **D.** Norming

2. A student takes the SAT, a test specifically designed not to measure what she has already learned, but instead to predict her future capacity and potential ability to succeed and learn new material in college. This type of test is called an:

- **A.** Achievement Test
- **B.** Norming Sample
- **C.** Reliability Test
- **D.** Aptitude Test

3. A psychologist wants to confirm that her new depression questionnaire truly measures the actual psychological concept of depression, rather than accidentally measuring something unrelated like general life stress or anxiety. Evaluating whether a test accurately captures the right underlying concept is called checking its:

- **A.** Split-Half Reliability
- **B.** Test-Retest Reliability
- **C.** Construct Validity
- **D.** Predictive Validity

4. A researcher measures intelligence by using a tape measure to record the exact circumference of each participant's head, getting the exact same precise measurement every single time she repeats the process on the same person. However, head circumference has absolutely nothing to do with actual intelligence. This test is best described as:

- **A.** Valid but Not Reliable
- **B.** Reliable but Not Valid
- **C.** Both Reliable and Valid
- **D.** Neither Reliable nor Valid

5. Researchers observe that the average IQ scores of entire populations have steadily risen generation after generation over the past century, likely due to improvements in nutrition, healthcare, and access to education rather than any actual genetic change in human intelligence. This observed global rise in IQ scores over time is called the:

- **A.** Stereotype Lift
- **B.** Flynn Effect
- **C.** Growth Mindset
- **D.** Testing Effect

6. Before taking a challenging exam, a student is reminded that people from his particular demographic group are stereotypically known for excelling on this type of test, and this awareness alone actually boosts his confidence and improves his performance. This phenomenon is called:

- **A.** Stereotype Threat
- **B.** A Growth Mindset
- **C.** Stereotype Lift
- **D.** The Flynn Effect

7. After failing his first calculus test, a student says, "I'm just naturally bad at math, and no amount of studying is ever going to change that, so there's no point in trying harder." This belief that intelligence and ability are fixed, static traits is called a:

- **A.** Fixed Mindset
- **B.** Growth Mindset
- **C.** Stereotype Lift
- **D.** Stereotype Threat

✅ Check Your Progress

Done with your notes? Head over to the Quiz to test what you've learned:
https://appsychreviewhub.com/quiz

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